

Raunak Kumar

raunak.kumar17@outlook.com
<https://raunakmr.github.io>

SUMMARY I take keen interest in machine learning, algorithms and competitive programming. My goal is to contribute to the advancement in machine learning research in order to develop novel solutions to real-life problems.

EDUCATION *B.Sc. Computer Science,* September 2013 - May 2018
University of British Columbia, Vancouver, BC

- Grade in Computer Science and Mathematics courses: 90.4%
- Selected coursework:
 - Graduate courses: Machine Learning, Randomized Big Data Algorithms, Probabilistic Graphical Models.
 - Advanced Algorithm Design and Analysis, Theory of Computation, Distributed Systems, Advanced Operating Systems.
 - Probability, Graph Theory, Real Analysis.

AWARDS (Nominated) Computing Research Association Outstanding Undergraduate Researcher Award (2018)
UBC Computer Science Teaching Assistant Award (2017)
NSERC Undergraduate Student Research Award (2017)
Science Scholar / Dean's Honors List (2013 - 2017)
J. Fred Muir Memorial Scholarship in Science (2014)
Trek Excellence Scholarship (2014) - Top 5% of undergraduates

WORK EXPERIENCE *NSERC Research Assistantship* May 2017 - Present
Computer Science Department, UBC
Supervisor: Mark Schmidt

- I am currently working on developing an improved variant of the expectation-maximization (EM) algorithm using techniques from numerical optimization. I am also investigating if deep neural networks can learn to recognize characters from a video stream without supervision.
- We view EM in a generalized surrogate optimization framework and provide the first convergence rate for non-convex functions in this framework and, consequently, also for EM. We have published a paper titled *Convergence Rate of Expectation-Maximization* at a workshop for optimization in machine learning at Neural Information Processing Systems (NIPS) 2017.
- I implemented random forest and neural network classification to detect dangerous gases based on sensor measurements for a company in Victoria, BC. I also presented low-dimensional visualization of the data using PCA and ISOMAP to aid the team in understanding patterns in the data.
- I assisted one of Prof. Schmidt's graduate students, Alireza Shafaei, on improving deep learning architectures for image segmentation. By incorporating the structure of the output space into the network, we obtained a 7% increase

in performance on the standard Cityscapes dataset on ResNet-38. Unfortunately, we didn't observe the same improvement in PSPNet, the state of the art network for image segmentation.

Software Engineering Intern
Google, Mountain View, CA

May 2016 - August 2016

- I wrote a distributed datastore trigger utility in C++ for the App and Product Infrastructure team. The utility provides an extensible way to maintain triggers, which are used to schedule and perform asynchronous actions. The utility was used to remove 1 billion triggers from storage.

TEACHING

Instructor / Coordinator
Computer Science Department, UBC

January 2017 - April 2017

- I co-taught CPSC 490: Problem Solving in Computer Science, focusing on practical applications of advanced algorithms and data structures.
- My responsibilities included lecturing a class of 15-30 students 3 times a week, and writing lecture notes and assignments.
- Website: www.ugrad.cs.ubc.ca/~cs490/2016W2/.

Teaching Assistant
Computer Science Department, UBC

September 2014 - Present

- I have worked as a TA for 9 terms for the following courses: Advanced Algorithm Design and Analysis, Machine Learning, Advanced Operating Systems, Intermediate Algorithm Design and Analysis, Introduction to Computer Systems, and Models of Computation. Usually, only graduate students are assigned to TA the first 3 courses.
- In January 2018, I will be a TA for CPSC 540, a graduate course on machine learning even though I am only an undergraduate student.
- My responsibilities include lecturing sections ranging from 15 - 100 students, holding office hours and grading. I have taught an entire lecture for two algorithms courses as well.

PUBLICATIONS

- **Raunak Kumar**, Mark Schmidt. *Convergence Rate of Expectation-Maximization*. NIPS OPT 2017.

PROJECTS

Programming Competitions

- UBC ACM programming team
 - Contestant: 2015 - Present.
 - Coach (Div. 2): 2017 - Present.
- ACM-ICPC PacNW Regionals
 - Contestant (Div. 1): 13th (2017), 15th (2016).
 - Contestant (Div. 2): 6th (2015).
 - Coach (Div. 2): 1st, 3rd (2017).
- Microsoft College Code Competition: 2nd (2017), 6th (2016).
- SFU Winter Programming Contest: 8th (2017).
- NAIPC (Open Division USA/Canada): 27th (2016).
- Calgary Collegiate Programming Competition: 3rd (2015, Div. 2).

Approximating Steiner Trees in the Semi-Streaming Model

- I presented a $2(1 + \epsilon) \left(2^{\frac{\log n}{\log \log n}} + 1 \right)$ -approximation for the weighted metric Steiner tree problem using $O(n \text{ polylog} n)$ space, as part of a course project for a graduate class on randomized big data algorithms. The paper can be found on my website.

Machine Learning Reading Group

- I present at the Machine Learning Reading Group and the Machine Learning Theory Reading Group. Past topics include learning theory, reinforcement learning and online learning. Slides can be found on my website.

Kernel

- I implemented basic functionality of a kernel using C and x86 assembly on the Bochs emulator. Features included memory management, process management, system calls, interrupt handling, signals and hardware interrupts.

Web Proxy Server

- I implemented a multi-threaded server application using C and Unix sockets to access HTTP content from websites, block blacklisted websites and cache results.

DNS Client

- I implemented a scaled down DNS resolver using Java and UDP datagrams to interact with various DNS servers to resolve domain names into IP addresses.

Dictionary Client

- I implemented a client side application using Java and TCP to connect with specified dictionary servers and retrieve word definitions.

insightUBC

- I implemented a backed web application using Typescript and Node.js to query UBC rooms and courses data.

TECHNOLOGY SKILLS

Programming Languages: C++, Python, C, Matlab.

Software: PyTorch, Numpy, Scikit-Learn, TensorFlow, Vim, Git.